

Education

<i>Doctor of Philosophy - Electrical and Computer Engineering (4.00/4.00)</i>	2024-Present
Purdue University - The Elmore Family School of Electrical and Computer Engineering	Indiana, US
Advisor: Prof. Abolfazl Hashemi	
<i>Master of Science - Electrical and Electronics Engineering (3.91/4.00)</i>	2021-2024
Bilkent University - Faculty of Engineering	Ankara, Turkey
Advisor: Prof. Süleyman Serdar Kozat	
<i>Bachelor of Science - Electrical and Electronics Engineering (3.61/4.00)</i>	2017-2021
Middle East Technical University - Faculty of Engineering	Ankara, Turkey

Research Background & Interests

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- Optimization (Constrained/Unconstrained Optimization Theory, Multi-Objective Optimization)
 - Trustworthy Deep Learning (Robustness, Privacy, Memorization)
 - Distributionally Robust Optimization (Invariant Learning, Subpopulation Shift, Domain Generalization)

Publications

Preprints

2. **A. Fazla**, E. Kaya, A. Upadhyay, A. Hashemi, “*Lower Bounds and Proximally Anchored SGD for Non-Convex Minimization Under Unbounded Variance*”, Under Review at NeurIPS 2026.
Available: <https://arxiv.org/abs/2604.16620>
1. A. Upadhyay, **A. Fazla**, A. Hashemi, “*Beyond Bounded Variance: Variance-Reduced Normalized Methods for Nonconvex Optimization under Blum-Gladyshev Noise*”, Under Review at NeurIPS 2026.
Available: <https://arxiv.org/abs/2605.15314>

Conference Papers

2. **A. Fazla**, A. Hashemi, “*Mitigating Spurious Correlations with Memorization-Guided Dataset De-Biasing Selection*”, Uncertainty in Artificial Intelligence, 2026.
Available: <https://arxiv.org/abs/2606.02830>
1. K. G. Ince, A. Köksal, **A. Fazla**, A. A. Alatan, “*Semi-Automatic Annotation for Object Detection*”, Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2021, pp. 1233-1239.
Available: <https://doi.org/10.1109/ICCVW54120.2021.00143>

Journal Papers

4. S. F. Tekin, **A. Fazla**, S. S. Kozat, “*Numerical Weather Forecasting using Convolutional-LSTM with Attention and Context Matcher Mechanisms*”, IEEE Transactions on Geoscience and Remote Sensing, 2024.
Available: <https://arxiv.org/abs/2102.00696>
Code: <https://github.com/sftekin/spatio-temporal-weather-forecasting>
3. **A. Fazla**, M. E. Aydin, S. S. Kozat, “*Time-Aware and Context-Sensitive Ensemble Learning for Sequential Data*”, IEEE Transactions on Artificial Intelligence, 2023.
Available: <https://doi.org/10.1109/TAI.2023.3319308>
Code: <https://github.com/ardafazla/context-time-aware-ensemble>
2. M. E. Aydin, **A. Fazla**, S. S. Kozat, “*Hybrid State Space-based Learning for Sequential Data Prediction with Joint Optimization*”, arXiv preprint arXiv:2309.10553, 2023.
Available: <https://arxiv.org/abs/2309.10553>
Code: <https://github.com/mustafaaydn/lstm-sx>

1. **A. Fazla**, M. E. Aydin, S. S. Kozat, “*Joint Optimization of Linear and Nonlinear Models for Sequential Regression*”, Digital Signal Processing, Elsevier, 2022.
Available: <https://doi.org/10.1016/j.dsp.2022.103802>
Code: <https://github.com/ardafazla/jointoptimization>

Professional Experiences

Purdue University, *Research Assistant (Advisor: Prof. Abolfazl Hashemi)* West Lafayette, IN

• *Stochastic Optimization* Aug 2025 - Present

- Developed PASTA and SAUCE, two stochastic optimization frameworks for nonconvex learning under realistic noise assumptions, covering the unbounded variance under the Blum-Gladyshev noise. Established theoretical convergence guarantees and demonstrated their relevance to modern deep learning settings where classical bounded-variance assumptions fail.
- Currently investigating the optimization and generalization behavior of deep neural networks, with a focus on sharpness-aware training dynamics and adaptive scheduling.

• *Trustworthy Deep Learning* Aug 2024 - Present

- Proposed a two-stage coreset selection algorithm with the dual purpose of eliminating spurious correlations and reducing dataset size. Deployed product achieved up to **27%** improvement over standard ERM utilizing only **10%** of the training data on benchmark datasets such as Waterbirds, CMNIST, MetaShift and UrbanCars, while also outperforming state-of-the-art invariant learning methods.
- Currently studying the training dynamics of deep learning models, with a focus on how loss landscape sharpness interacts with common training interventions and influences memorization and generalization, particularly in out-of-distribution settings.

Databoss Security & Analytics Inc., *Data Scientist* Ankara, Turkey

• *Hourly Wind Energy Prediction* Jun 2023 - May 2024

- Predicted the hourly energy data of multiple wind turbines in Turkey and Europe, showing chaotic and nonstationary behavior in multiple regions.
- Developed a large-scale ML framework consisting of various adaptive feature construction and selection methods with models such as deep-learning (NN, ConvLSTM), machine learning (LightGBM, RandomForest), statistical (SARIMAX) models and state-of-the-art ensembling techniques. The deployed product increased the short-term forecasting accuracy by an average of **21%** and long-term forecasting accuracy by an average of **8%**.

• *Natural Gas Demand Prediction* Aug 2021 - May 2023

- Predicted the daily data of international natural gas demand in Turkey, consisting of nonstationary multivariate time series from multiple sources.
- Developed a large-scale ML framework specifically tailored towards a certain customer profile, where the predictions are obtained for the given data based on key indicators that are automatically extracted from the data itself and/or the user-given side information data. The constructed framework was deployed to various natural gas production companies.

METU Center for Image Analysis (OGAM), *Machine Learning Engineer* Ankara, Turkey

• *UAV Small Object Detection and Tracking* Jun 2020 - Jul 2021

- Developed an application in Python environment for real-time small target tracking by leveraging temporal data derived from the output of object detectors, with a superior performance over the widely-known YOLOv3 algorithm by **6%**.
- Constructed a novel multi-hypothesis tracking algorithm with an annotation tool for the semi-automatic correction of mislabeled UAV data. The deployed product decreased the total mislabeled data in various datasets by an average of **13%**.

Academic Awards

Elmore Family School of Electrical and Computer Engineering, Purdue University 2024-2029

- Awarded the highly competitive and prestigious Elmore Fellowship Award for my graduate studies, given only to the most outstanding applicants.

Turk Telekom & Information and Communication Technologies Authority 2023-2024

- Awarded 5G and Beyond Joint Graduate Support Programme, a merit-based fellowship of monthly stipend during M.Sc.

Scientific and Technological Research Council of Turkey 2021-2023

- Awarded Directorate of Science Fellowships and Grant Programme, a merit-based monthly stipend and accommodation support during M.Sc.
- Received the **373rd** rank among 2M high school graduates in the University Entrance Examination.
- Received the **11th** rank among 0.2M university graduates in ALES (National GRE).

Academic Duties

Reviewer Duties

- **Conferences:** ICLR '25-'26, NIPS '26, AISTATS '26, AAAI '25
- **Journals:** IEEE TAI '25, IEEE IOT '24-25

Teaching Assistantship at Bilkent University, Electrical and Electronics Engineering 2021-2024

- EEE321 Signals and Systems, EEE202 Circuit Theory, MATH255 Probability and Statistics

Software Skills

- **Python:** Professional research and industrial experience based on machine learning. High knowledge and experience in libraries such as *PyTorch*, *Tensorflow*, *Pandas*, *Numpy* and *Scikit-Learn*. I have experience with Docker and Git via industrial projects.
- **MATLAB:** Professional research experience in signal processing and computer vision. Used for projects during courses, research and internships.
- **R:** Intermediate level experience, employed during research based on time-series forecasting using statistical models.
- **C/C++:** Intermediate experience in various course projects.

Languages

Turkish: Native

English: Conversationally Fluent [TOEFL IBT 108/120 (October 19, 2022)]